

Isolation of nuclei from tissue culture cells

Risk assessment

- Work with human-derived material or transgenic cell lines (BSL-2)
- ▷ Wear gloves, safety glasses, lab coat
- Collect and dispose waste after inactivation as REGULATED MEDICAL WASTE



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This is a bench card. Full protocol available online.

Preparation of cells

- Phosphate-buffered saline, pH 7.4
- R0090 100 × Inhibitor cocktail, 1 mL

- (1.) Start with 1×10^6 – 1×10^9 cells. When working with frozen cell pellets, thaw on ice. Approximate the packed cell volume; this is 1 vol. Typically, 1×10^9 cells amount to 1–2 mL cell volume.

Critical: Keep buffer volumes to a minimum to maintain high protein concentrations.

- (2.) Wash cells twice with 5 vol ice-cold PBS with inhibitors. Try to obtain a suspension free of visible clumps. Collect the cells at $400 \times g$ for 5 min at 4 °C.



⌚ 10 min



A > Detergent-free isolation of nuclei

- Dounce homogenizer, loose pestle
- R0091 Swelling buffer, 10 mL

- (1.) Resuspend the cell pellet in 5 vol ice-cold $1 \times$ hypotonic swelling buffer with inhibitors. Ensure the suspension is free of clumps. Incubate for 10 min on ice to swell the cells.

⌚ 10 min

- (2.) Collect the cells by centrifugation at $400 \times g$ for 10 min at 4 °C, discard the supernatant.

⌚ 10 min

- (3.) Resuspend in 2 vol (at least 0.4 mL) swelling buffer. Transfer to a pre-chilled Dounce homogenizer.



Critical: Do not vortex! Great care must be exercised throughout this procedure. Repeated pipetting and centrifugation will result in compromised nuclei which are very susceptible to leaking and disruption.



- (4.) Using a type A (or “loose”) pestle, lyse the cells by 10–15 strokes. Apply even pressure and deliberate strokes for uniform lysis. Avoid the formation of bubbles and foam. Repeat until lysis is complete.



Quality assurance: Monitor lysis by removing 5–10 µL lysate. Mix with an equal volume of 0.4% Trypan blue; view under a microscope. Stop homogenization when 95% of cells lost their cytoplasmic membrane and nuclei still remain intact.



- (5.) Transfer to a centrifuge tube, pellet broken cells and nuclei at $400 \times g$ for 5 min at 4 °C. Keep the supernatant as cytoplasmic fraction if desired.



Quality assurance: The nuclear pellet should be white as opposed to the off-white/beige cellular pellet.



- (6.) Wash once in 1.5 vol hypotonic swelling buffer to remove cell membranes.

- (7.) Resuspend the nuclei in the desired buffer for downstream experiments.

B > Isolation of nuclei in isosmotic sucrose

- R0092 Sucrose buffer, 10 mL
- R0057 10% Triton™ X-100

- (1.) Wash the cell pellet twice in 2–5 vol ice-cold isosmotic sucrose buffer with inhibitors.

- (2.) Resuspend in 1–2 vol ice-cold sucrose buffer with inhibitors. Ensure the suspension is free of clumps. Admix an equal volume of sucrose buffer containing 0.6% Triton X-100. Incubate for 10 min on ice.

⌚ 10 min



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Critical: If no single cell suspension is obtained at this point, cell lysis will not be homogenous. Avoid shearing membranes and organelles by pipetting very gently. ←

- (3.) Pellet at $400 \times g$ for 5 min at 4 °C. If desired, keep the supernatant as cytoplasmic fraction.
- (4.) Wash twice with 2–5 vol sucrose buffer to remove cell membranes.
- (5.) Resuspend the nuclei in the desired buffer for downstream experiments.

Storage of isolated nuclei

- (1.) Resuspend the nuclei in isosmotic sucrose buffer; snap-freeze in liquid nitrogen or a dry ice–ethanol bath. Store at –80 °C for up to two years. Do not refreeze.

🔗 Recipe (available online) 🛠 Troubleshooting (available online) 📖 Notes (available online)

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